Asif Ahmed Joy

Ph.D. Candidate, Information Systems
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Professional Summary

Ph.D. candidate in Information Systems specializing in augmented or extended or mixed reality (AR/XR/MR) simulation for healthcare and caregiver training. Prototypes AR/XR/MR systems with embodied virtual human agents on Microsoft HoloLens 2 under a National Science Foundation (NSF) award, and evaluates them through mixed methods research pairing quantitative usability, technology acceptance, and social presence studies with qualitative interviews and thematic analysis. Grounded in human computer interaction (HCI), user experience research (UX/UXR), and artificial intelligence (AI) and machine learning (ML). Author of nine peer reviewed conference publications, seven as first author, at venues including IEEE ISMAR, IEEE VR, and ACM IVA. Four years of undergraduate teaching at NJIT as instructor of record and teaching assistant.

Education

- **Ph.D. in Information Systems** *2021 – Present*
  New Jersey Institute of Technology, Newark, New Jersey, USA Expected Graduation: December 2026
  – GPA: 3.46/4.00 Advisor: Dr. Salam Daher

- **B.Sc. in Computer Science & Engineering** *2020*
  Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh
  – Thesis: Dimensionality Reduction (using PCA, LDA, mRMR) for Hyperspectral Image Classification (using SVM, KNN, ANN)

Research Experience

Doctoral Research

- **Doctoral Thesis: Aware Intelligent Virtual Humans in Mixed Reality** *In Progress*
  New Jersey Institute of Technology, Newark, New Jersey, USA *Advisor: Dr. Salam Daher*
  – Automating spatial awareness in embodied virtual humans in AR/XR/MR so that context appropriate non verbal behaviors are generated without a human operator in the loop
  – Benchmarking the automated system against a human controlled baseline on non verbal response accuracy and system latency
  – Designing and conducting a user study evaluating perceived accuracy, latency, and realism of the virtual human's automated non verbal responses

- **Embodied Virtual Human Simulation for Caregiver Training** *Jan 2022 – Present*
  Department of Informatics, New Jersey Institute of Technology Unity 3D, C#, HoloLens 2, MRTK
  – Student team lead on the NSF Future of Work at the Human Technology Frontier (FW-HTF) collaborative project (Awards #2222661, #2222662, #2222663) spanning NJIT, the University of Central Florida, and the University of Delaware
  – Prototyped an interactive AR/XR/MR simulation featuring a spatially aware embodied geriatric virtual human patient for caregiver communication and soft skills training
  – Ran quantitative user studies measuring system usability, technology acceptance, social presence, and caregiver soft skills including empathy, attitudes, and self efficacy
  – Ran qualitative studies by interviewing professional caregivers and analyzed the interview data using thematic analysis to derive findings that informed subsequent simulation design
  – First authored four papers published and presented at IEEE ISMAR, IEEE VR, and ACM IVA

- **Smart Healthcare Assistant on Amazon Alexa** *2022*
  New Jersey Institute of Technology, Newark, New Jersey, USA Python, AWS, Google Cloud Platform
  – Built and deployed an Amazon Alexa skill on AWS that delivers healthcare information to older adults, backed by a dynamic central database

Undergraduate Research

- **Dimensionality Reduction for Hyperspectral Image Classification** *2018 – 2020*
  Undergraduate Thesis Research, RUET, Rajshahi, Bangladesh Matlab, Python, scikit-learn
  – Developed hybrid feature selection and feature extraction pipelines (PCA, LDA, mRMR, nMI) paired with SVM, KNN, and ANN classifiers for multi band satellite imagery
  – Produced three peer reviewed IEEE and ACM conference publications from this work

Industry Experience

- **Machine Learning Engineering Intern** *Jan 2021 – Mar 2021*
  Pioneer Alpha, Dhaka, Bangladesh Python, RASA, TensorFlow, Keras, PyTorch
  – Built conversational AI natural language understanding models on the RASA framework using transfer learning

Teaching Experience

- **Instructor of Record, IT 201: Information Design Techniques** *Spring 2025*
  Department of Informatics, New Jersey Institute of Technology, Newark, New Jersey, USA
  – Sole instructor of record for an undergraduate section of 26 students, without teaching assistant or grader support

- **Graduate Teaching Assistant and Tutor** *Sep 2021 – Present*
  Department of Informatics, New Jersey Institute of Technology, Newark, New Jersey, USA
  – IT 201: Information Design Techniques • Tutored undergraduate students on information design concepts and course assignments
  – IT 420: Computer Systems and Networks • Graded projects, homework, and course assignments
  – IS 350: Computers, Society and Ethics • Supported instruction and facilitated discussions on ethical, legal, and social issues in computing

Training and Certifications

- CITI Program: Human Subjects Research Ethics and Compliance Training *2022, renewed 2026*

Professional Memberships

- Institute of Electrical and Electronics Engineers (IEEE)
- Association for Computing Machinery (ACM)

Peer Reviewed Conference Publications

Author name appears as **Ahmmed, A.** or **Joy, A. A.** Full list available on Google Scholar.

Doctoral Research

1. **Ahmmed, A.**, Thiamwong, L., Barmaki, R. L., Ng, B. P., Ancis, J. R., & Daher, S. (2026). SimPathy: Simulation Design and Evaluation of an Aware Embodied Patient in Mixed Reality for Caregivers' Soft Skills Training. *1st International Workshop on XR for Behavioral Health (BehavXR), 2026 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR Adjunct)*, Bari, Italy. **(Accepted; to be presented October 2026.)**

2. **Ahmmed, A.**, Butts, E., Naeiji, K., Thiamwong, L., & Daher, S. (2024). System Usability and Technology Acceptance of a Geriatric Embodied Virtual Human Simulation in Augmented Reality. *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 594–603.

3. **Ahmmed, A.**, Butts, E., Naeiji, K., Thiamwong, L., Ancis, J., & Daher, S. (2024). Aware Intelligent Virtual Agent's Effect on Social Presence and Empathy of Caregivers. *Proceedings of the 24th ACM International Conference on Intelligent Virtual Agents (IVA)*, 1–5.

4. **Ahmmed, A.**, Butts, E., Naeiji, K., Thiamwong, L., & Daher, S. (2024). Simulation of an Aware Geriatric Virtual Human in Mixed Reality. *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, 1162–1163.

5. Kiafar, B., Daher, S., Sharmin, S., **Ahmmed, A.**, Thiamwong, L., & Barmaki, R. L. (2024). Analyzing Nursing Assistant Attitudes Towards Geriatric Caregiving Using Epistemic Network Analysis. *International Conference on Quantitative Ethnography (ICQE)*, 187–201. Springer.

6. Kiafar, B., Daher, S., **Ahmmed, A.**, Thiamwong, L., & Barmaki, R. L. (2023). A Quantitative Ethnographic Examination to Improve the Quality of Training for Caregivers for a Simulated Immersive Virtual Reality Setting. *Fifth International Conference on Quantitative Ethnography (ICQE)*, Vol. 8.

Undergraduate Research

7. **Joy, A. A.**, Hasan, M. A. M., & Sayeed, A. (2020). An Improved Class-wise Principal Component Analysis Based Feature Extraction Framework for Hyperspectral Image Classification. *Proceedings of the International Conference on Computing Advancements (ICCA)*, ACM, Article 48, 1–6.

8. **Joy, A. A.**, & Hasan, M. A. M. (2019). A Hybrid Approach of Feature Selection and Feature Extraction for Hyperspectral Image Classification. *2019 International Conference on Computer, Communication, Chemical, Materials and Electronic Engineering (IC4ME2)*, IEEE, 1–4.

9. **Joy, A. A.**, Hasan, M. A. M., & Hossain, M. A. (2019). A Comparison of Supervised and Unsupervised Dimension Reduction Methods for Hyperspectral Image Classification. *2019 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, IEEE, 1–6.